

AAS

Automated Accounting System

Application Flow & User Guide

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A comprehensive guide to the AAS system architecture, user workflows, and integration patterns.

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1. System Overview

What is AAS?

AAS (Automated Accounting System) is a comprehensive procurement, inventory, and finance management platform designed for multi-branch businesses. It streamlines order creation, vendor management, inventory tracking, and invoice generation while integrating with ERPNext.

Architecture

```
Angular UI (Frontend)
|
Spring Boot Middleware (Business Logic)
|
ERPNext (Data Layer)
```

Key Features

- Dual-mode order creation: Image-based (OCR) or catalog-based
- Vendor PDF parsing with AI-assisted template generation
- Automated invoice creation with GST support
- Role-based access control via feature flags
- Real-time dashboards: Orders, Vendors, Branches, Stock
- Multi-branch operational views with ledger tracking
- Payment allocation and receivables management

2. Authentication & Login Flow

Login Process

User navigates to /login -> Enters credentials -> POST /api/auth/login -> JWT token returned -> Stored in localStorage -> Redirected to dashboard

Token Storage

- `aas_auth_token`: JWT bearer token
- `aas_auth_role`: User role (admin, shop, etc)
- `aas_auth_features`: Array of feature flags
- `aas_auth_home_route`: Default route for user's role

Security Guards

- `authGuard`: Checks if token exists. If missing, redirects to /login
- `featureGuard`: Calls GET /api/me to fetch user profile. Validates feature permissions. If unauthorized, bounces to user's homeRoute

Logout

User clicks Logout -> AuthTokenService clears all localStorage keys -> Redirects to /login

3. Application Shell & Navigation

Layout

Header: Company logo, branch info, user profile

Sidebar: Feature-filtered navigation tree (modules user has access to)

Main Content: Currently selected module

Dynamic Sidebar

On init, `buildNavSections()` filters the full nav tree to only show modules matching user's feature flags. This ensures role-based visibility.

Navigation Sections

- Home: Dashboard
- Procure: Orders, Vendors
- Inventory: Stock, Items
- Finance: Bills, Payments
- Master Data: Vendors, Branches, Categories, Items
- Reports: Various reports
- Operations: Vendor Ops, Branch Ops

4. Dashboard

Overview

Snapshot of operations for current calendar month with multiple KPI cards.

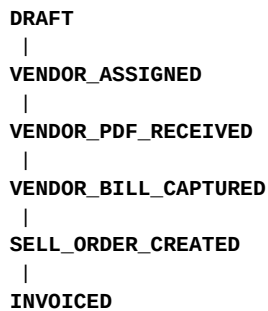
KPIs Displayed

- Order Status Breakdown: Count by status
- Sales Summary: Invoice count, total revenue
- Stock Snapshot: Item count, total quantity
- Billing by Vendor: Top vendors by amount
- Billing by Branch: Top branches by amount
- Vendor Operations: 6 KPIs (total vendors, pending orders, etc)
- Branch Operations: 6 KPIs (total branches, pending orders, etc)

5. Orders Workflow

(Current Development Branch: order_workflow_branch)

Order Status Pipeline



Order Creation: Mode 1 (Image-Based)

- Upload photos of branch order sheets
- Backend uses OCR/AI to parse items
- Best for: Existing physical order sheets

Order Creation: Mode 2 (Catalog-Based)

- Browse items from catalog, filtered by category
- Items pre-populated with vendor pricing
- Submit structured item list
- Best for: Known items, precise quantities

Order Management

- Assign Vendor: Link order to supplier
- Upload Vendor PDF: Scan/upload vendor invoice
- Parse PDF: Backend AI extracts line items
- Capture Bill: Enter vendor bill total, reference, date
- Sell Preview: View margin before selling
- Create Sell Order: Creates Purchase Invoice in ERPNext

6. Vendor Operations

Summary View

Table showing all vendors with pending order counts, PDF status, bill capture status, pending amount, last activity, and ledger balance.

Per-Vendor Drill-Down

- KPIs: Pending orders, PDF status, bill status
- Template Status: Invoice parsing template setup
- Billing Summary: Bills captured, outstanding amount
- Exceptions: Mismatches, parse failures, overdue PDFs
- Orders Table: Status and details per order
- Ledger: Double-entry with running balance

Settlement States

- Settled: Ledger balance ~0
- Open: Outstanding balance exists
- Overdue: Unpaid invoices or overdue PDFs

CSV export available for ledger analysis.

7. Branch Operations

Overview

Branch-centric operational view tracking order pipelines, receivables, and payment health.

Summary Metrics

- Pending Orders
- Awaiting Vendor Assignment
- Open Receivables
- Payment Collection Rate
- Ledger Balance
- Last Activity

Per-Branch View

- KPIs: Pending orders, receivables, payment rate
- Billing Summary: Invoices, payments, balance
- Exceptions: Unassigned orders, overdue invoices
- Orders Table: Full order details per branch
- Ledger: Transaction history with running balance

8. Bills & Invoicing

Invoice Creation: From Order

- Select order -> Pre-fills customer, items, quantities
- Toggle GST, adjust rates, set rounding
- Submit -> Creates Sales Invoice in ERPNext

Invoice Creation: Manual

- Select customer and company
- Manually add line items
- Apply GST and taxes
- Submit -> Creates Sales Invoice

GST Handling

- Auto-creates Item Tax Template (AAS GST X%)
- Creates GST account under Duties & Taxes
- Applies taxes_and_charges template

Payment Recording

- Optionally link to invoice (auto-fills outstanding)
- Enter payment amount
- Calculates balance_after_payment and surplus
- Submit -> Creates Payment Entry in ERPNext

Invoice Deletion

- Finds all linked Payment Entries
- Cancels and deletes payment entries
- Cancels and deletes invoice
- Graceful handling of GL entry retention

9. Master Data

Vendors (Suppliers)

- Supplier records and management
- Invoice template setup for PDF parsing
- Payment terms and contact details

Branches (Shops/Customers)

- Branch records and management
- Credit days for invoicing
- Location, contact, default currency

Categories

- Item grouping for easier browsing
- Used in order creation filters

Items (Catalog)

- Item code, name, category, vendor
- Cost and sell rates
- GST rate per item
- Vendor-specific pricing

10. Stock Monitoring

Item Tracking

- Item ID, code, name, vendor
- Current quantity on hand
- Reorder threshold (per-item)
- Status: Low (below threshold) or OK

Summary Panel

- Total Items: count of all items
- Total Quantity: sum of quantities
- Low Stock Count: items below threshold
- Vendor Breakdown: items grouped by vendor

Threshold Management

Thresholds stored in browser localStorage (key: aas_stock_thresholds). Per-device, not synced across browsers. Items with null threshold are not alerted.

11. API Reference

Authentication

- POST /api/auth/login (username, password)
- GET /api/me (user profile + features)

Orders

- GET /api/orders (list with filters)
- POST /api/orders (create from items)
- POST /api/orders/branch-image (create from image)
- POST /api/orders/{id}/assign-vendor
- POST /api/orders/{id}/vendor-pdf (upload PDF)
- POST /api/orders/{id}/vendor-bill (capture bill)
- GET /api/orders/{id}/sell-preview
- POST /api/orders/{id}/sell-order

Invoices & Payments

- GET /api/invoices (list invoices)
- POST /api/invoices (create invoice)
- GET /api/invoices/{id}/pdf (download PDF)
- DELETE /api/invoices/{id}
- POST /api/payments (record payment)

Vendor Operations

- GET /api/vendor-ops/summary
- GET /api/vendor-ops/{id}
- GET /api/vendor-ops/{id}/orders
- GET /api/vendor-ops/{id}/ledger

Branch Operations

- GET /api/branch-ops/summary
- GET /api/branch-ops/{id}
- GET /api/branch-ops/{id}/orders
- GET /api/branch-ops/{id}/ledger

Common Workflows

Workflow 1: Create Order to Invoice to Payment

- Orders -> Create Order (upload image or select items)
- Orders -> Manage (assign vendor, upload PDF, capture bill)
- Bills -> Create Invoice (from order snapshot)
- Bills -> Payment (record payment with allocation)

Workflow 2: Monitor Vendor Health

- Operations -> Vendor Ops (view summary of all vendors)
- Click vendor (drill-down view with KPIs and exceptions)
- Review pending orders, PDF status, ledger balance
- Export ledger to CSV for analysis

Workflow 3: Track Branch Receivables

- Operations -> Branch Ops (view all branches)
- Click branch (drill-down with KPIs and invoices)
- Review payment collection rate and overdue invoices
- Export branch ledger to CSV

End of Documentation

This comprehensive guide covers the AAS system architecture, user workflows, and API reference. For questions, contact the development team.

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